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ITECH3102 –Data Visualization
Week 4

Lab Exercises –SVG, D3

ID: _____Name: _____

In this week's projects, we discuss some fundamentals of Scalable Vector Graphics (SVG) and D3 (Data-Driven Documents or D3.js.

Project 1: Scalable Vector Graphics (SVG)

1. What is SVG?

Scalable Vector Graphics (SVG) is an XML-based vector image format for defining two-dimensional graphics, having support for interactivity and animation.

SVG images are defined in a vector graphics format and stored in XML text files. SVG images can thus be scaled in size without loss of quality, and SVG files can be searched, indexed, scripted, and compressed. The XML text files can be created and edited with text editors or vector graphics editors, and are rendered by the most-used web browsers.

D3 is most useful when used to generate and manipulate visuals as Scalable Vector Graphics (SVG). Using SVG is more reliable, visually consistent, and faster.

2. How to use inline SVG images

SVG images can be written directly into the HTML document using the
<svg>

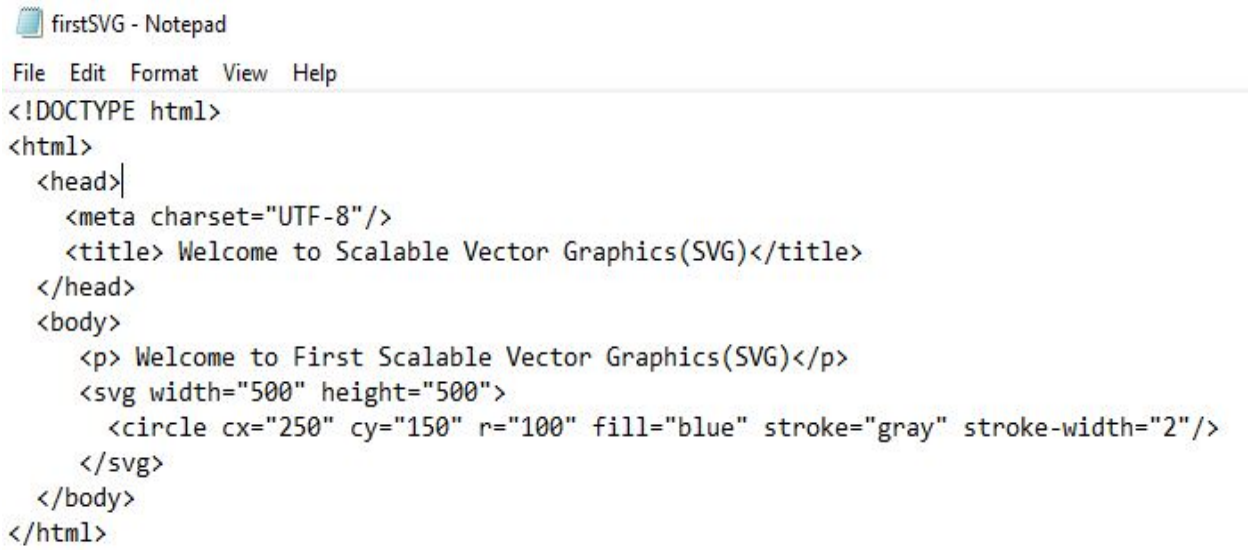
</svg> tag.

To do this, write codes in a text editor or in Developer tools or your preferred an integrated development environment (IDE), copy the code, and paste it inside the <body> element in your HTML document.

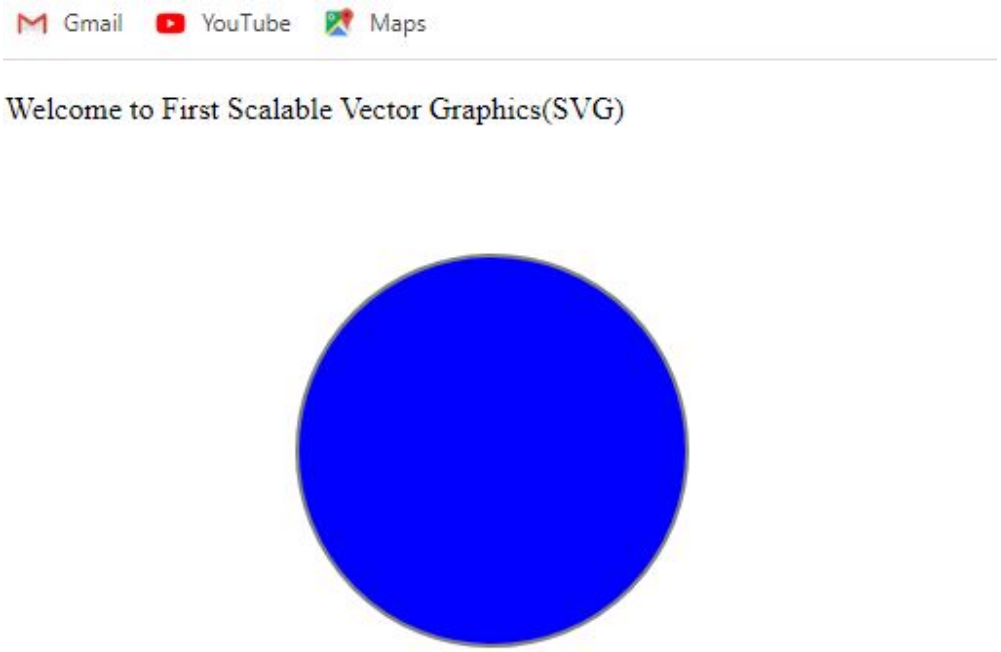
<body>
// put your SVG code here.
</body>

Please note that if you add SVG codes into html file by using Developer tools, you need to press Ctrl + S key to save the updated html file to your local computer.
.

Type the following codes in text editor like Notepad, and save it.



After saving firstSVG.html in your computer, right click the html file and select Google Chrome. You'll see the following





We explain the above codes later.

3. The SVG Element and Simple Shapes

SVG element

SVG is a text-based image format. Each SVG image is defined using markup code similar to HTML, and SVG code can be included directly within any HTML document,

or inserted dynamically into the DOM. **Every current web browser** supports SVG. SVG is XML-based, so you'll notice that elements that don't have a closing tag must be self-closing. For example:

```
<element></element> <!-- Uses closing tag -->
<element/>          <!-- Self-closing tag -->
```

It's good to specify width and height values. If you don't specify these, the SVG will take up as much room as it can within its enclosing element. In the previous example:

Line 9

```
<svg width="500" height="500">
</svg>
```

SVG coordinates system

SVG coordinates system is shown in the following figure:



The SVG coordinates system

In the SVG coordinates system, 0,0 is the top-left corner of the drawing space. Increasing x values move to the right, while increasing y values move down.

Simple shapes

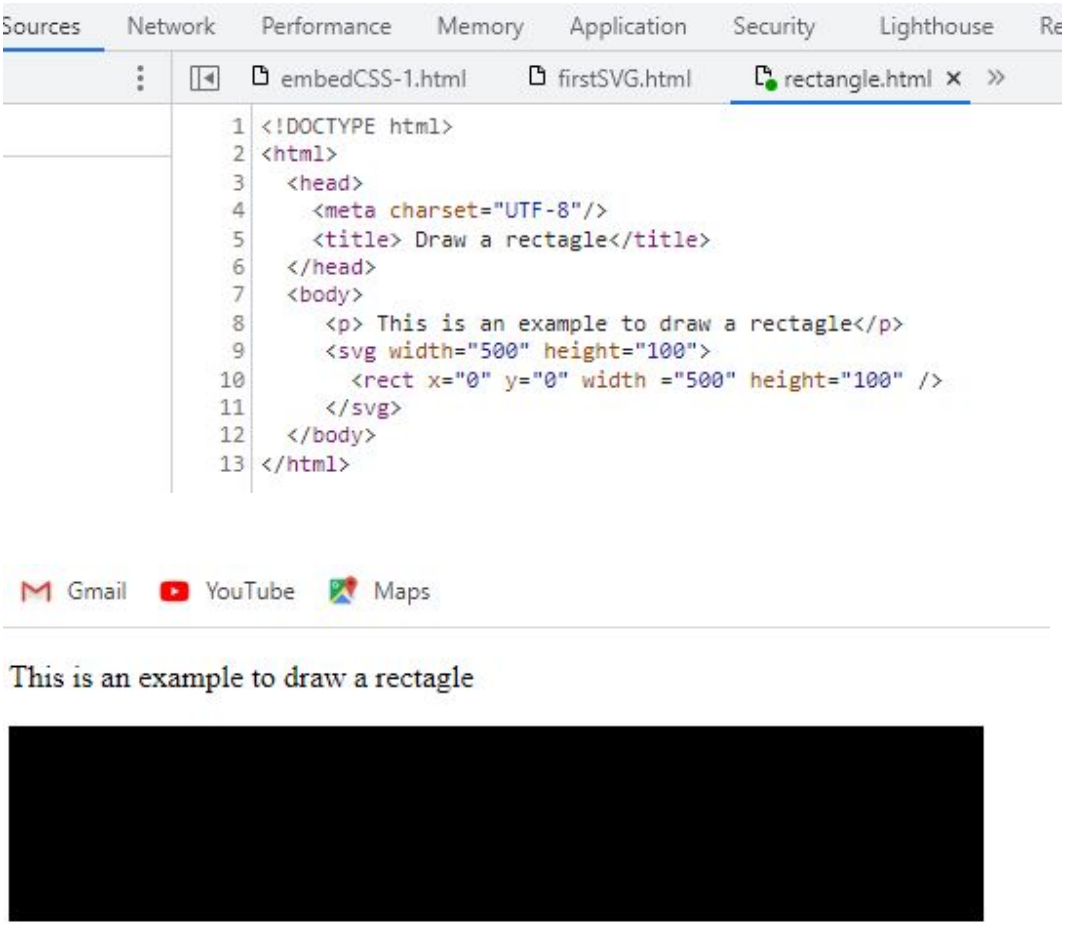
You can include a number of visual elements between svg tags to draw a shape. These visual elements include rect**, circle, ellipse, line, text, and path.**

Rectangle

`rect` draws a rectangle. Use x and y to specify the coordinates of the upper-left corner, and width and height to specify the dimensions. This rectangle fills the entire space of our SVG, as shown in the following picture.

```
<rect x="0" y="0" width="500" height="100"/>
```

Type the following codes in text editor or Developer tools and run it.



Circle

In the following example, we'll draw a circle:



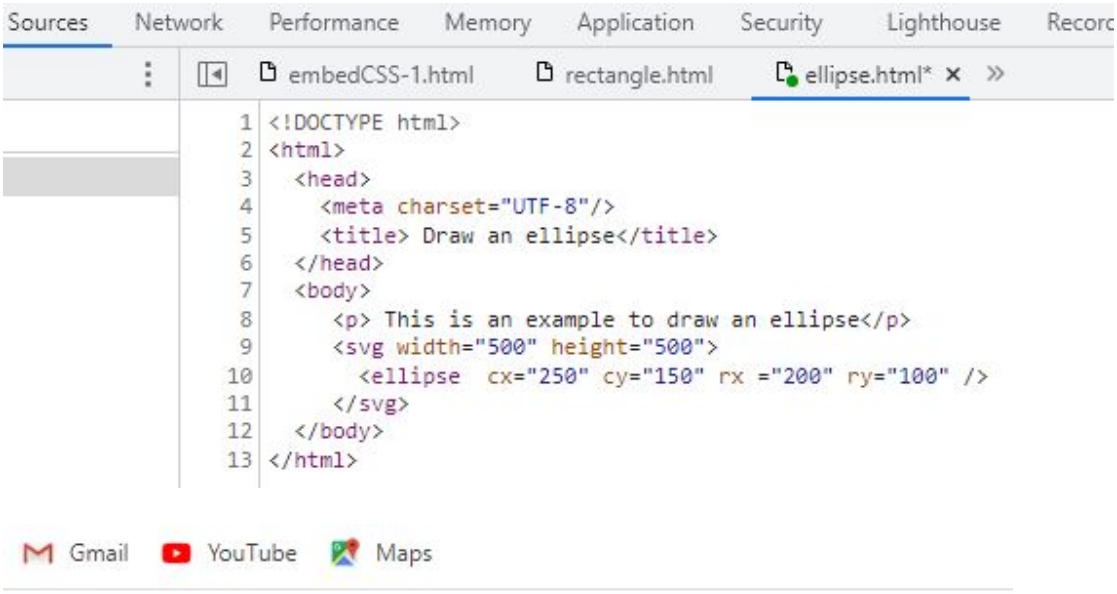
In the above codes, line 10

<circle cx="250" cy="150" r="100" fill="blue" stroke="gray" stroke-width="2"/>

element **circle** draws a circle. Use **cx** and **cy** to specify the coordinates of the *center*, and **r** to specify the radius. This circle is centred in the middle of our 500-pixel-wide SVG because its cx (“center-x”) value is 250.

Ellipse

Type the following codes in text editor or Developer tools and run it.



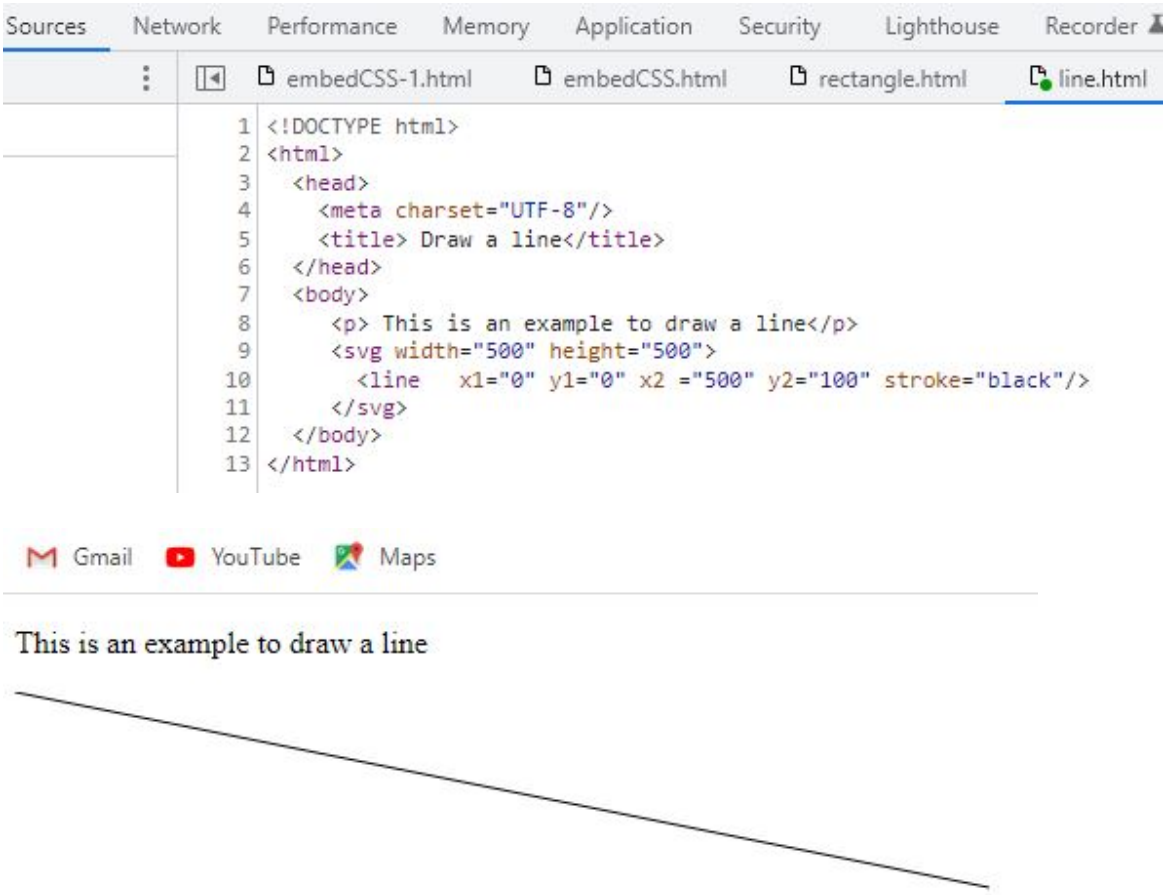
In the above codes, line 10

<ellipse cx="250" cy="150" rx ="200" ry="100" />

ellipse draws an ellipse. Here ellipse uses separate radius values rx and ry for each axis.

SVG Line

Type the following codes in text editor or Developer tools and run it.



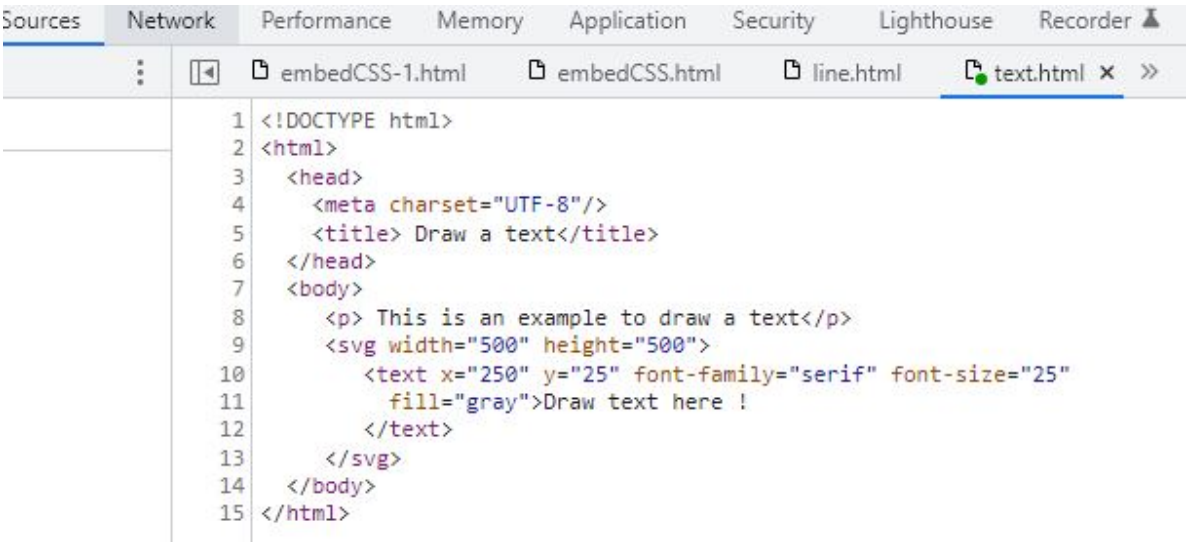
In the above codes, line 10

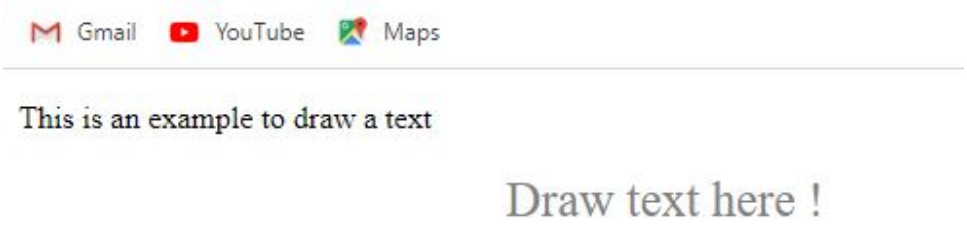
<line x1="0" y1="0" x2 ="500" y2="100" stroke="black"/>

line draws a line. Use x1 and y1 to specify the coordinates of one end of the line, and x2 and y2 to specify the coordinates of the other end. A stroke color must be specified for the line to be visible:

SVG text

Type the following codes in text editor or Developer tools and run it.





In the above codes, line 10, 11 and 12

```
<text x="250" y="25" font-family="serif" font-size="25"
        fill="gray">Draw text here !
 </text>
```

render text by using element text. Use x to specify the position of the left edge, and y to specify the vertical position of the type's baseline.

Project 2: Styling SVG elements, layering and transparency

1. Styling SVG elements

SVG's default style is a black fill with no stroke. If you want anything else, you'll have to apply styles to your elements. Common SVG properties are as follows:

fill
A color value. Just as with CSS, colors can be specified as named colors, hex values, or RGB or RGBA values.

stroke
A color value.

stroke-width
A numeric measurement (typically in pixels).

opacity
A numeric value between 0.0 (completely transparent) and 1.0 (completely opaque).

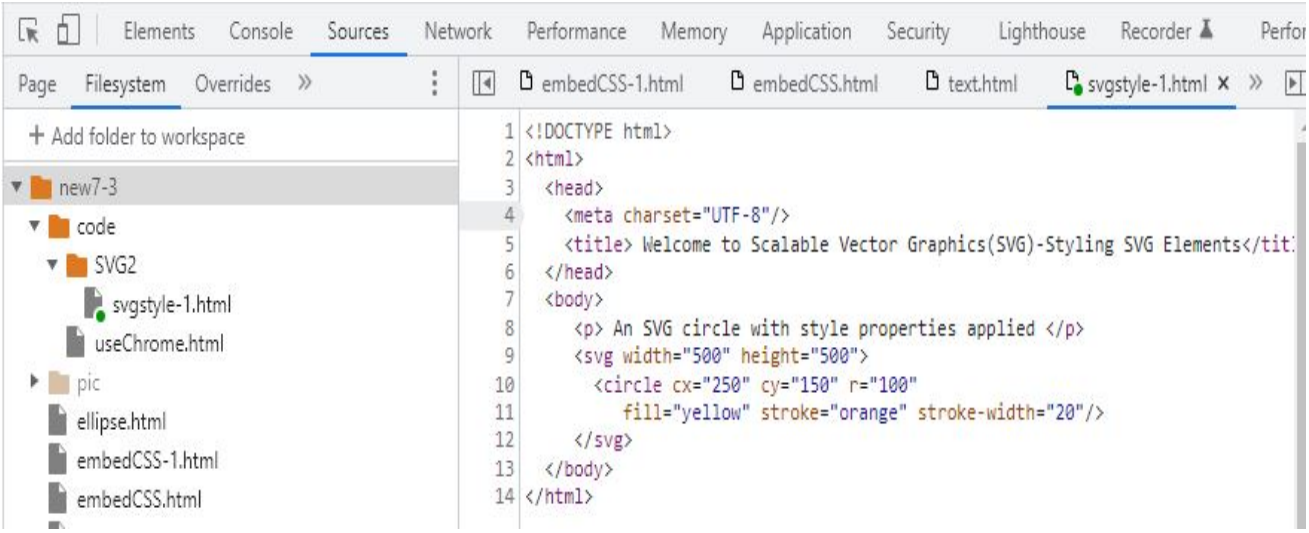
With text, you can also use these properties, which work just like in CSS:

font-family

font-size

In another parallel to CSS, there are two ways to apply styles to an SVG element: either directly (inline) as an attribute of the element, or with a CSS style rule.

Type the following codes in a text editor or Developer tools and run it.



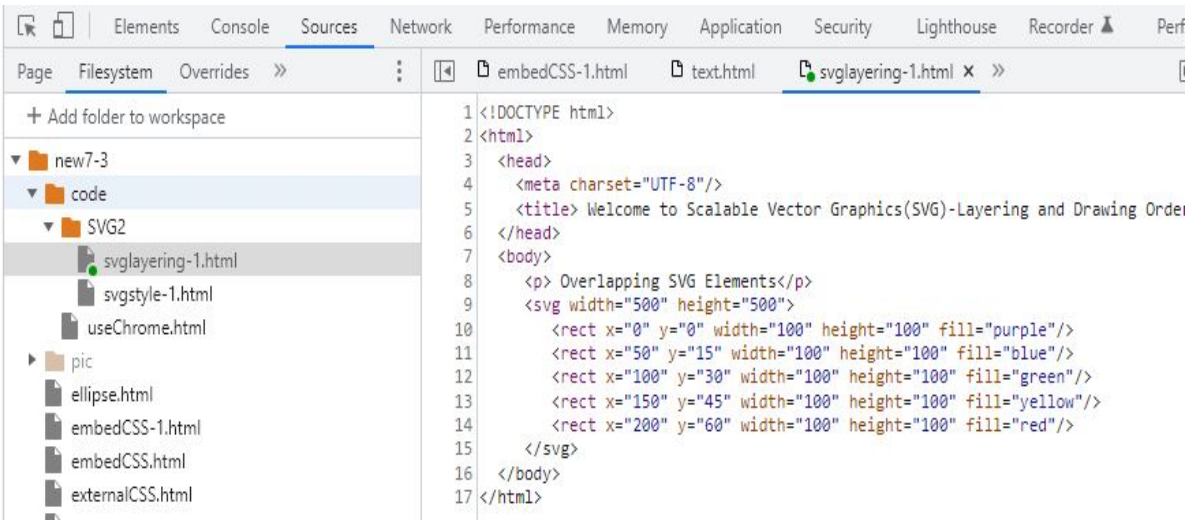


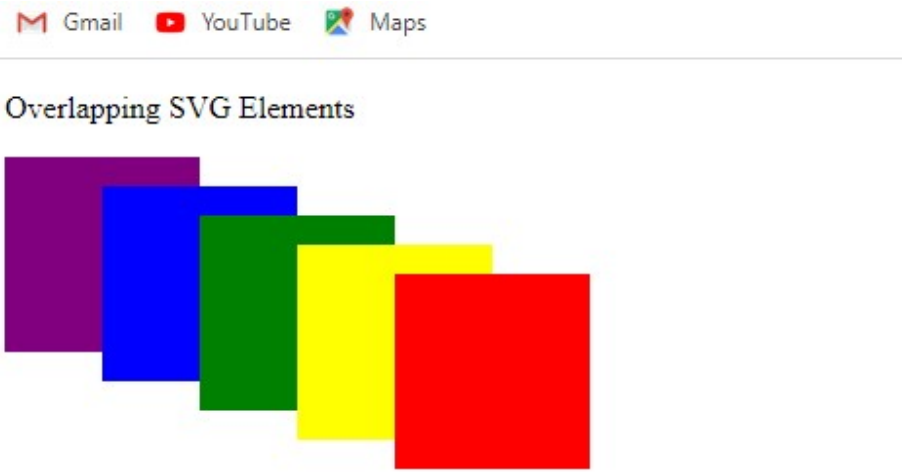
In the above codes, some style properties are applied directly to a circle as attributes in line 10 and 11.

2. Layering and Drawing Order

There are no “layers” in SVG and no real concept of depth. SVG does not support CSS’s z-index property, so shapes can be arranged only within the two-dimensional x/y plane.

Type the following codes in a text editor or Developer tools and run it.



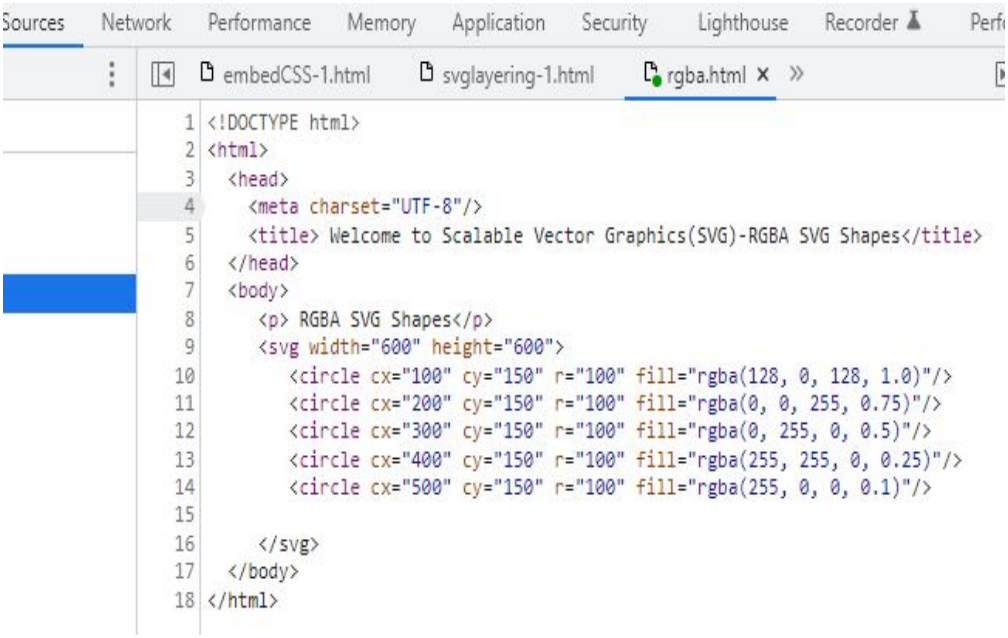


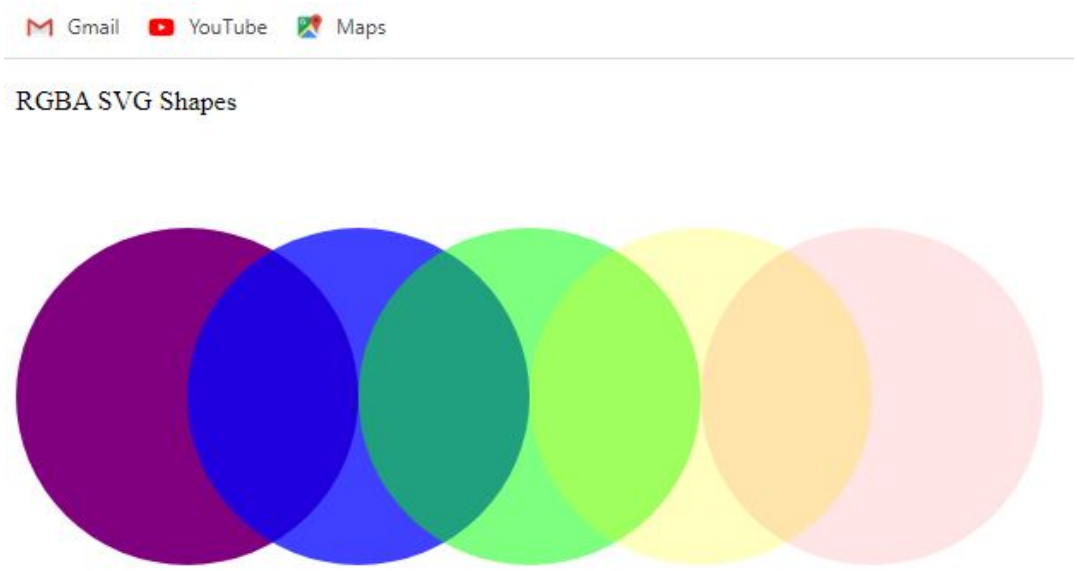
In the above codes, the order in which elements are coded determines their depth order. In the Figure above, the purple square appears first in the code, so it is rendered first. Then, the blue square is rendered “on top” of the purple one, then the green square on top of that, and so on.

3. Transparency

Transparency can be useful when elements in your visualization overlap but must remain visible, or you want to deemphasize some elements while highlighting others, as shown in the following Figure.

Type the following codes in a text editor or Developer tools and run it.





There are two ways to apply transparency: use an RGB color with alpha, or set an opacity value.

In the above codes, you can use rgba () anywhere you specify a color, such as with fill or stroke. rgba() expects three values between 0 and 255 for red, green, and blue, plus an alpha (transparency) value between 0.0 and 1.0, which are shown through lines 10 to l4.

Note that with rgba(), transparency is applied to the fill and stroke colors independently.

Project 3. Setting up a D3 development environment

When you start a D3-powered data visualization project, the first thing you need to do is to set up a working development environment where you can create, edit and test your codes. In this project, we first create a project folder and download the latest version of D3.js, then install Python and finally set up a local HTTP server.

1. Create a project folder and download the latest version of D3.js

We first create a project folder where two Main components: d3.js and index.html are stored there.

(1) Creating a project folder

In my computer, Python will be stored within the folder: C:\Users\Mike. Then I create a project folder whose structure look something like this:

C:\Users\Mike\ITECH3102
d3.js
index.html

(2) Downloading D3

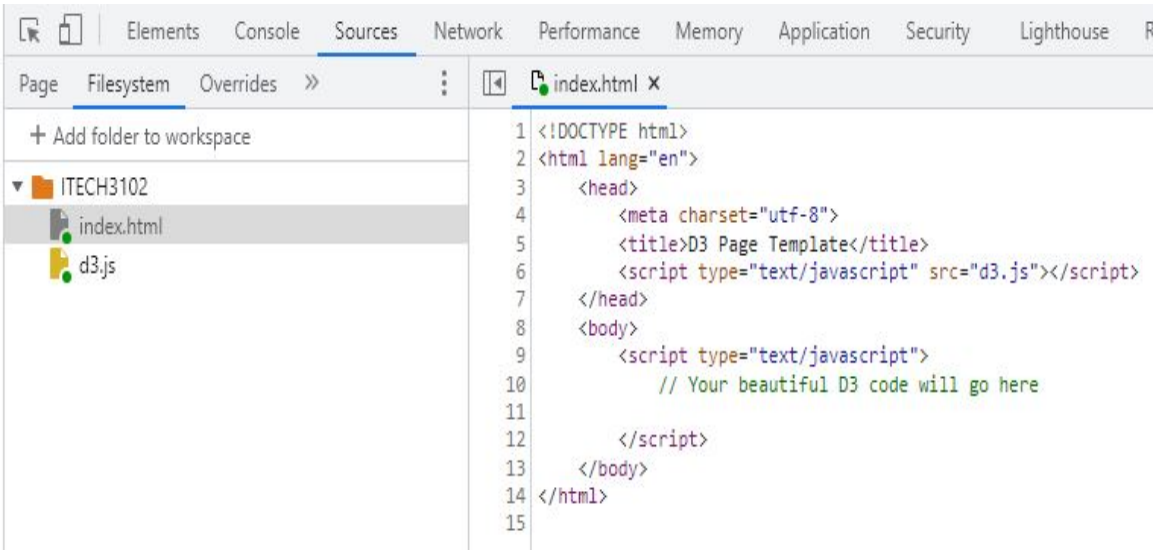
You can download D3 from Moodle. For development purpose, it is recommended that you use the d3.js file. Therefore, move the file d3.js into your project folder.

(3) Referencing D3

Once downloading D3.js, create a simple HTML file named **index.html** and place it in the same folder as D3.js.

Type the following codes in a text editor and place it in the same folder as D3.js:





In the above codes in the template:

In line 4, meta tag identifies the encoding for this file as utf-8, which is needed to ensure that the browser can parse D3's functions and data properly.

In line 6, the first script tag sets the reference to d3.js. You should edit this file path as needed if you decide to locate d3.js somewhere other than the project-folder directory.

In line 9, the second script tag, in the body, is where you will soon key in all your beautiful code.

Done! Your D3 template files and folders are all set up. If you want to start a new project, you can **make a copy of this template**, which makes it easy for you to start a new project.

2. Install Python

We now install Python. We use free Anaconda Python distribution in this course because it's easy to install on Windows and support the latest version of Python (3.x). The software includes most of Python, visualization and data science libraries you need.

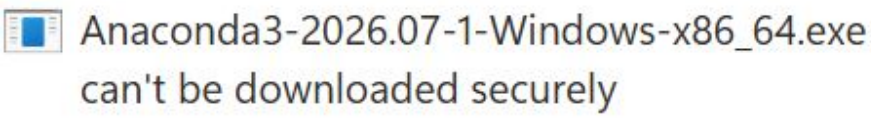
First of all, you need to install Anaconda on your PC.

(1) Go to website:
https://www.anaconda.com/download/

to download the Python 3.x Anaconda installer for Windows:

Before downloading software Anaconda Distribution, you need to Sign Up with your email. Once signing up, you can download Anaconda Distribution.

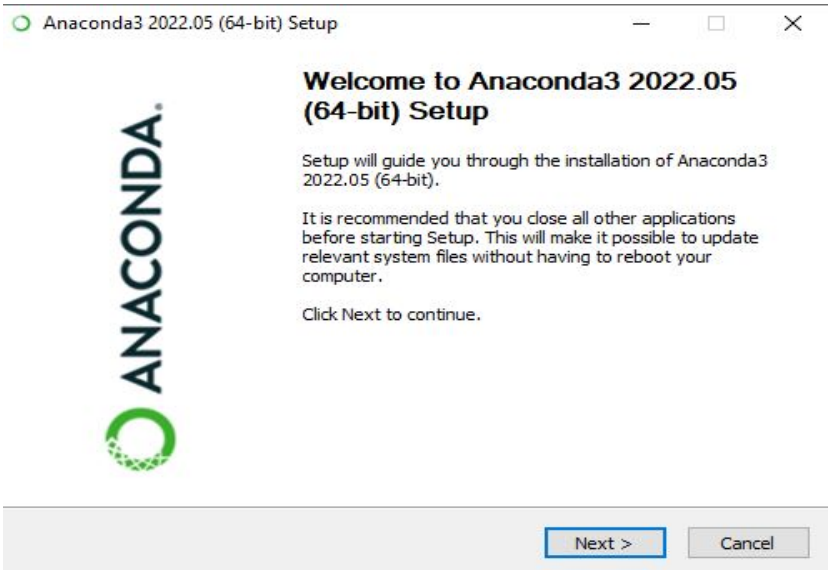
Please note that when downloading Anaconda Distribution, you may receive the following prompt: "This software cannot be downloaded securely."

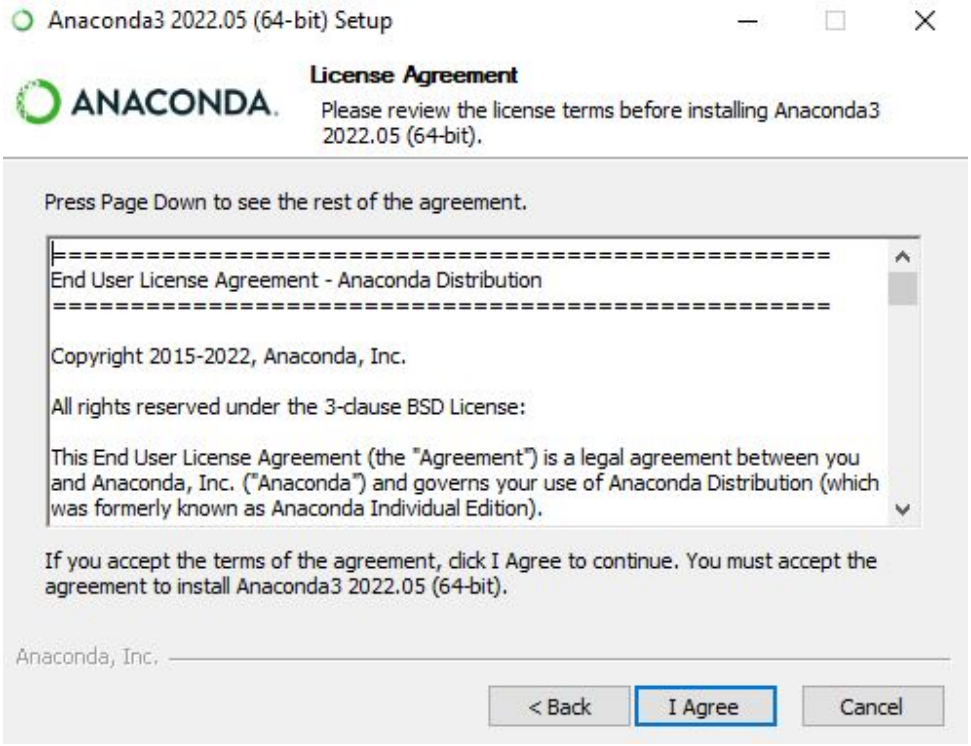


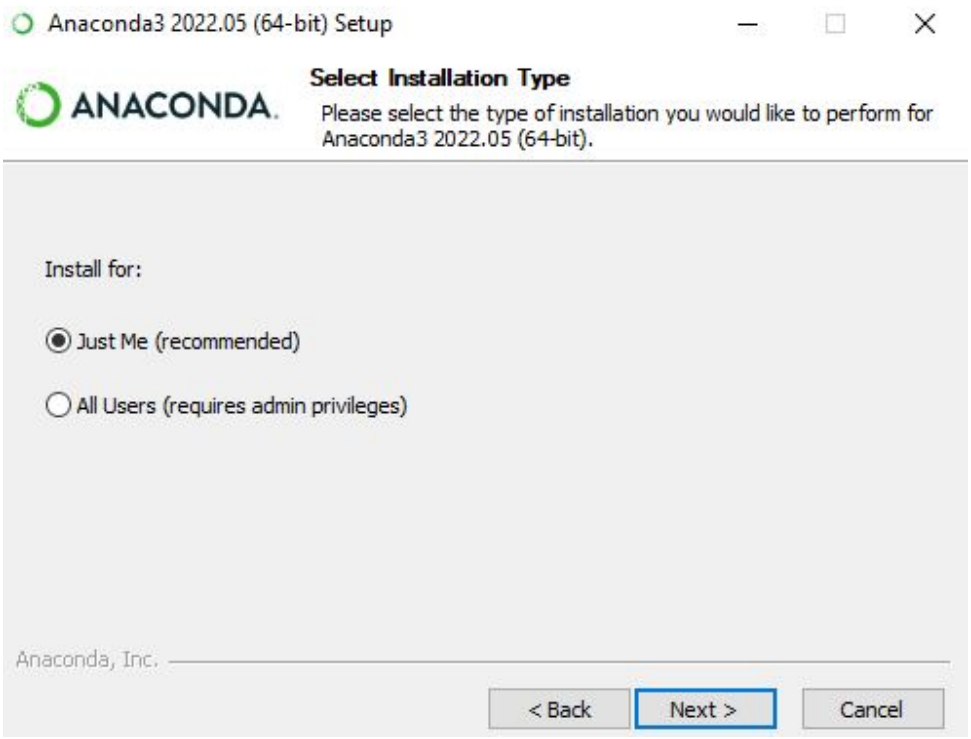
If this happens, I suggest you download the **Miniconda Installer** software, which includes conda, Python and related packages. Therefore, you can complete this week's experiments after downloading **Minconda Installer**.

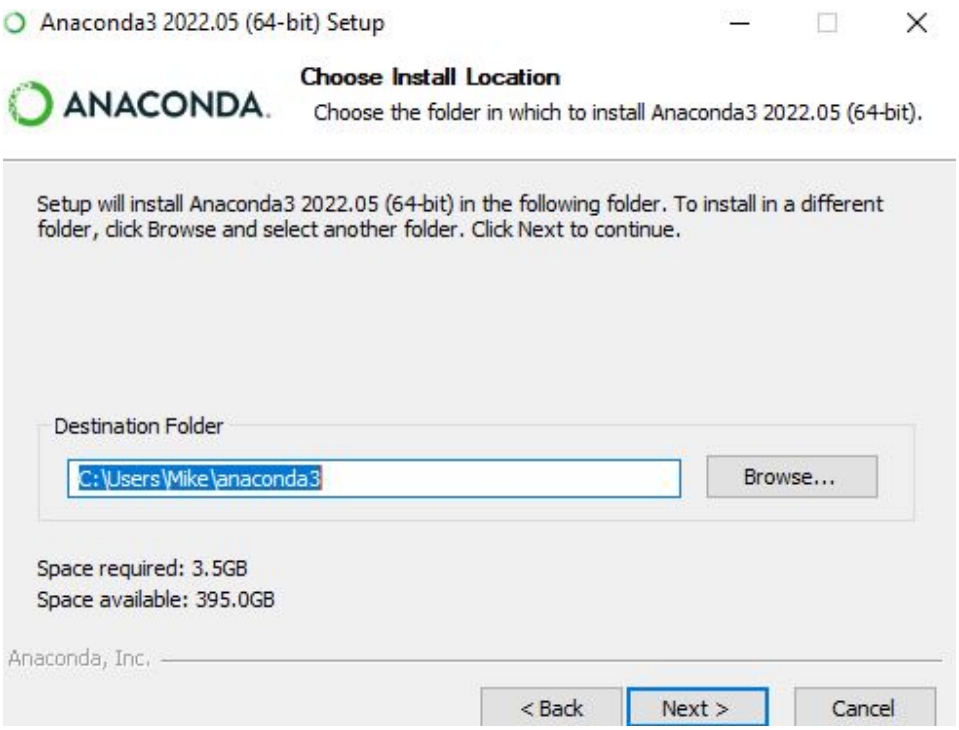


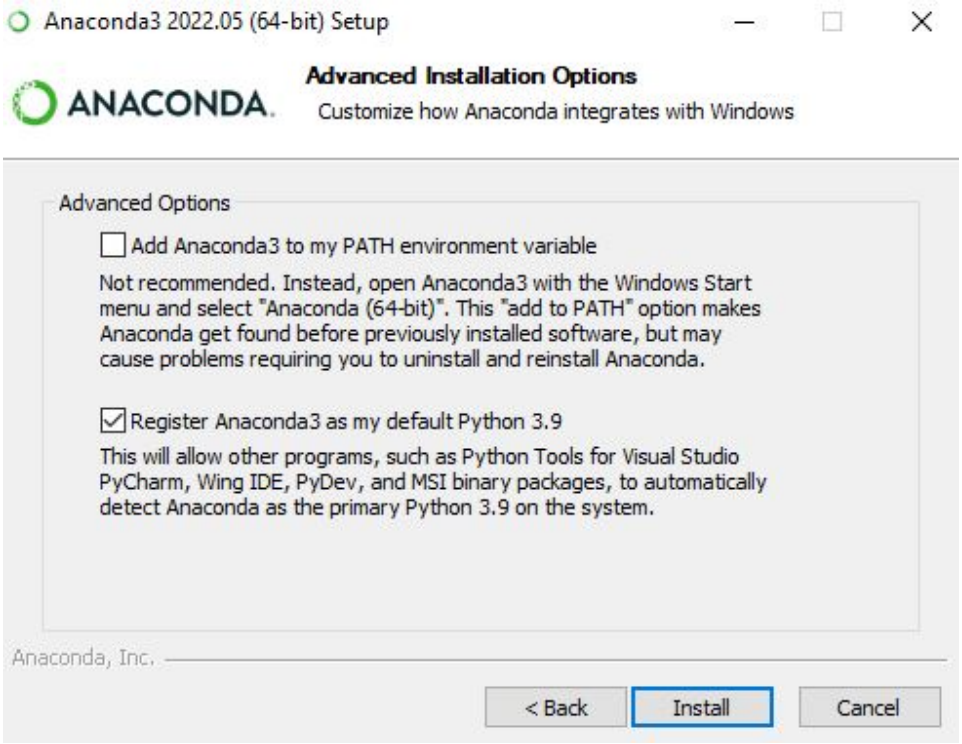
(2) After clicking Download button, save downloaded Anaconda in your local computer (in my computer, I save Anaconda within the folder C:\Users\Mike). Then click Anaconda3-2022.05-Windows-x86_64.exe to install the software.

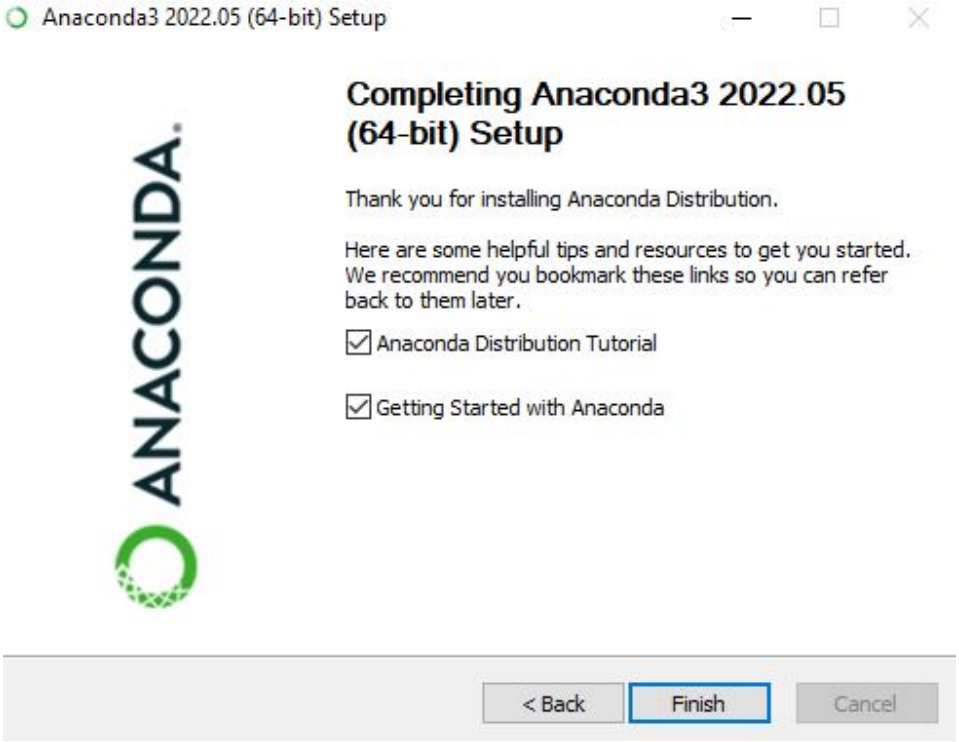



Anaconda3 2022.05 (64-bit) Setup
ANACONDA.
License Agreement
Please review the license terms before installing Anaconda3 2022.05 (64-bit).
Press Page Down to see the rest of the agreement.
==========
End User License Agreement - Anaconda Distribution
==========
Copyright 2015-2022, Anaconda, Inc.
All rights reserved under the 3-clause BSD License:
This End User License Agreement (the "Agreement") is a legal agreement between you and Anaconda, Inc. ("Anaconda") and governs your use of Anaconda Distribution (which was formerly known as Anaconda Individual Edition).
If you accept the terms of the agreement, click I Agree to continue. You must accept the agreement to install Anaconda3 2022.05 (64-bit).
Anaconda, Inc.
< Back
I Agree
Cancel


Anaconda3 2022.05 (64-bit) Setup
ANACONDA.
Select Installation Type
Please select the type of installation you would like to perform for Anaconda3 2022.05 (64-bit).
Install for:
Just Me (recommended)
All Users (requires admin privileges)
Anaconda, Inc.
< Back
Next >
Cancel

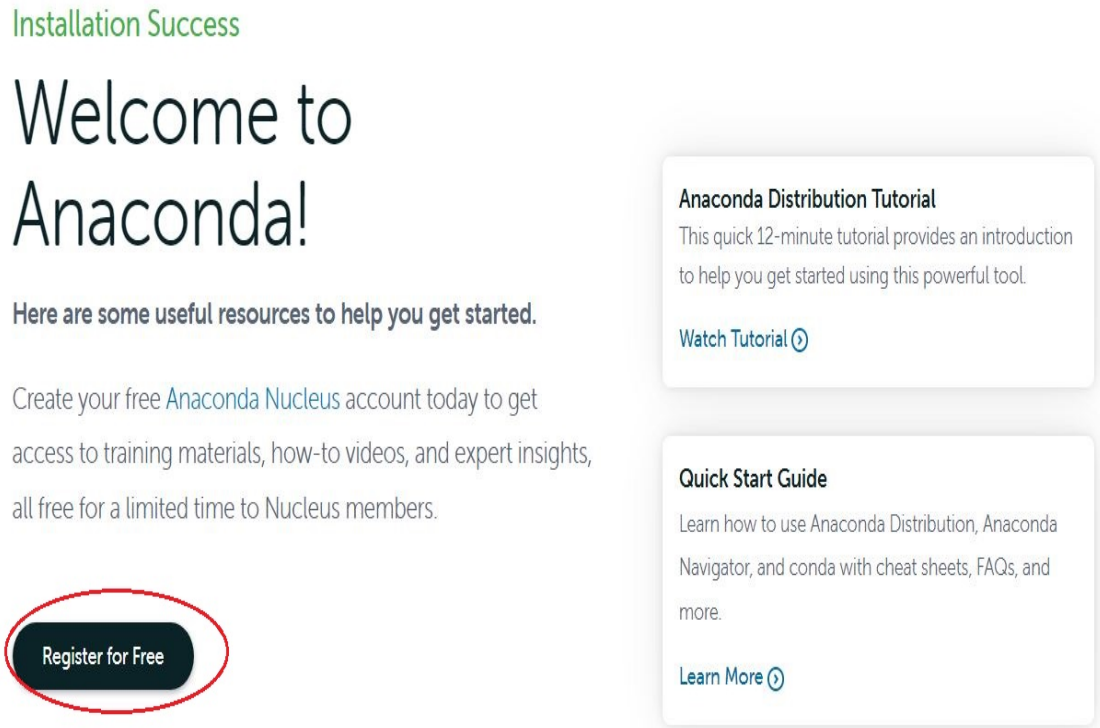

Anaconda3 2022.05 (64-bit) Setup
ANACONDA.
Choose Install Location
Choose the folder in which to install Anaconda3 2022.05 (64-bit).
Setup will install Anaconda3 2022.05 (64-bit) in the following folder. To install in a different folder, click Browse and select another folder. Click Next to continue.
Destination Folder
C:\Users\Mike\anaconda3
Browse...
Space required: 3.5GB
Space available: 395.0GB
Anaconda, Inc.
< Back
Next >
Cancel


Anaconda3 2022.05 (64-bit) Setup
ANACONDA.
Advanced Installation Options
Customize how Anaconda integrates with Windows
Advanced Options
Add Anaconda3 to my PATH environment variable
Not recommended. Instead, open Anaconda3 with the Windows Start menu and select "Anaconda (64-bit)". This "add to PATH" option makes Anaconda get found before previously installed software, but may cause problems requiring you to uninstall and reinstall Anaconda.
Register Anaconda3 as my default Python 3.9
This will allow other programs, such as Python Tools for Visual Studio PyCharm, Wing IDE, PyDev, and MSI binary packages, to automatically detect Anaconda as the primary Python 3.9 on the system.
Anaconda, Inc.
< Back
Install
Cancel

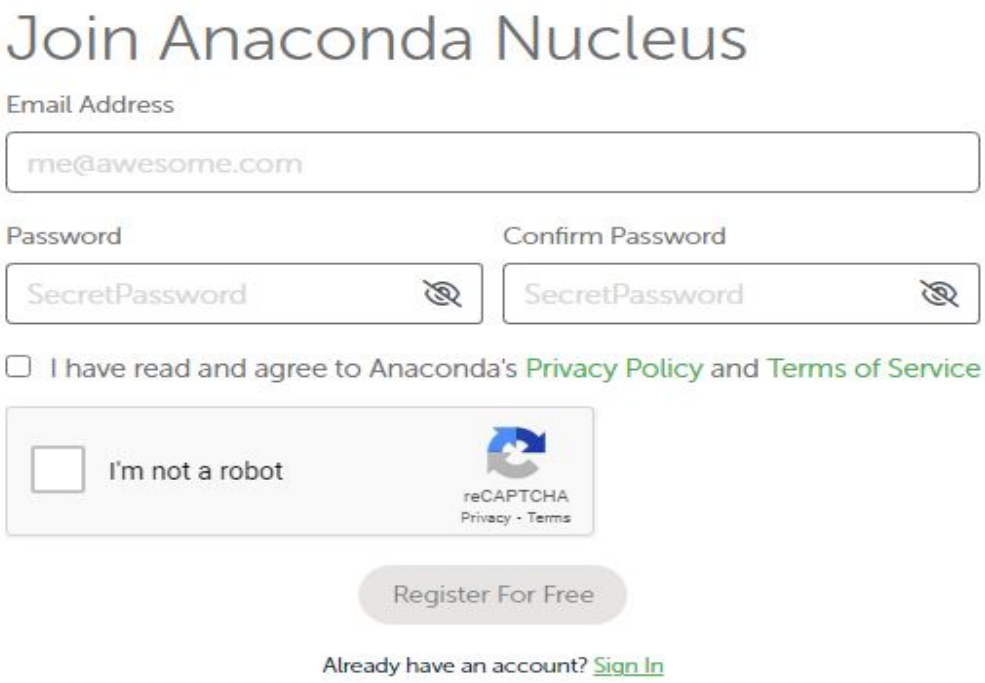

Anaconda3 2022.05 (64-bit) Setup
ANACONDA.
Completing Anaconda3 2022.05 (64-bit) Setup
Thank you for installing Anaconda Distribution.
Here are some helpful tips and resources to get you started. We recommend you bookmark these links so you can refer back to them later.
Anaconda Distribution Tutorial
Getting Started with Anaconda
< Back
Finish
Cancel

(3) Register for free software

After successfully installing the software, you need to register for free software and then press Register for Free button.

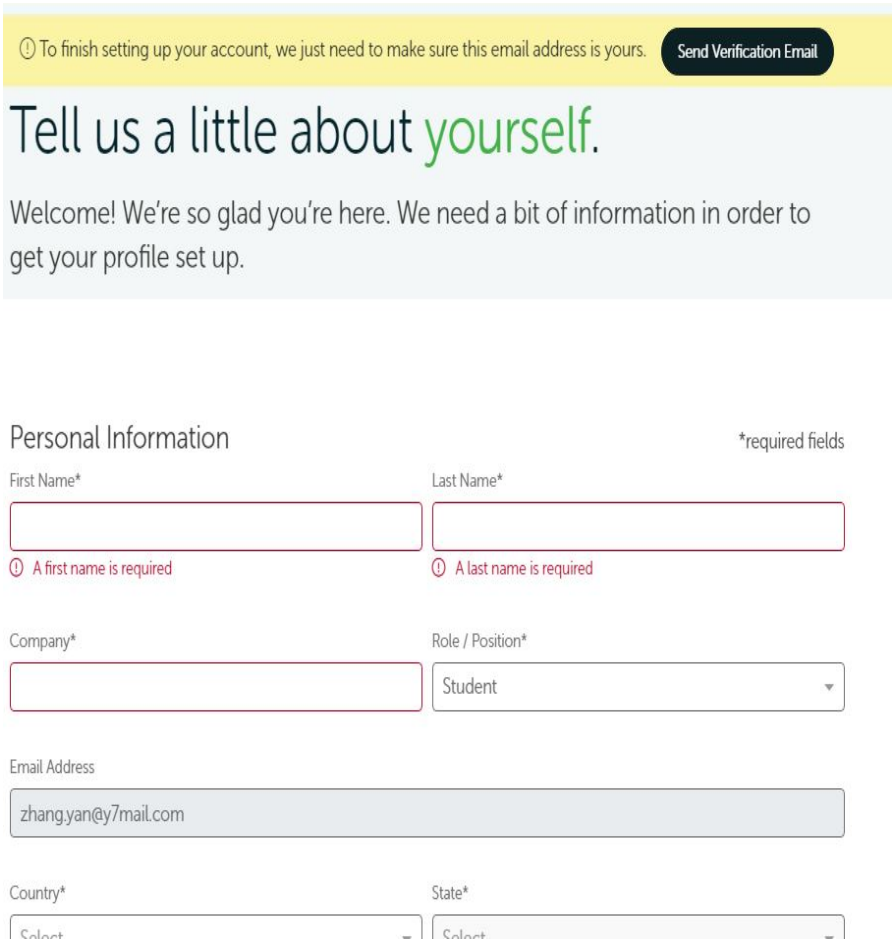


You need to provide related information to register for free software and then press Register for Free button.



(4) Complete your profile

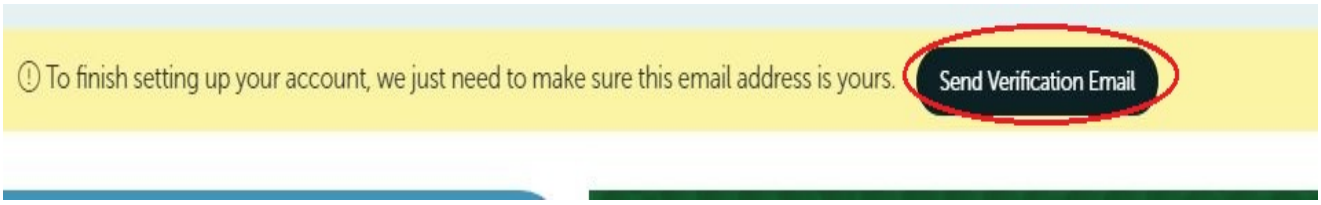
After clicking Register For Free, you need to provide your profile.



After providing your profile, make sure that you save your information.

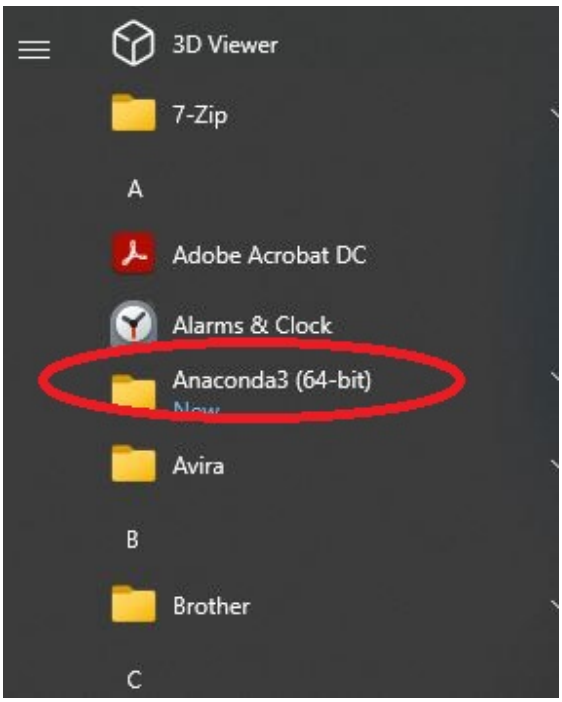
(5) Verify your email

After completing and saving your profile, you need to click **Send Verification Email** and finally go to your email account to verify your email.

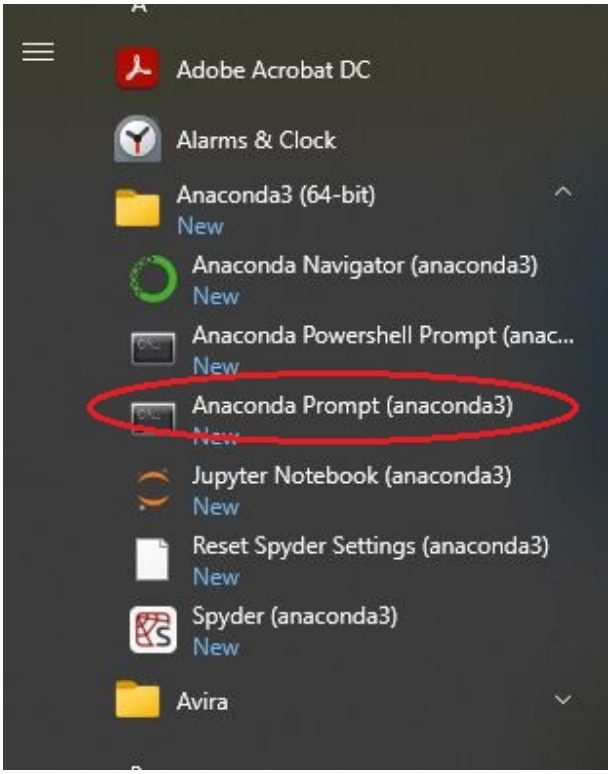


(6) Updating Anaconda

You should ensure that Anaconda is up to date. From your start menu in your computer, you can find the Anacoda3.

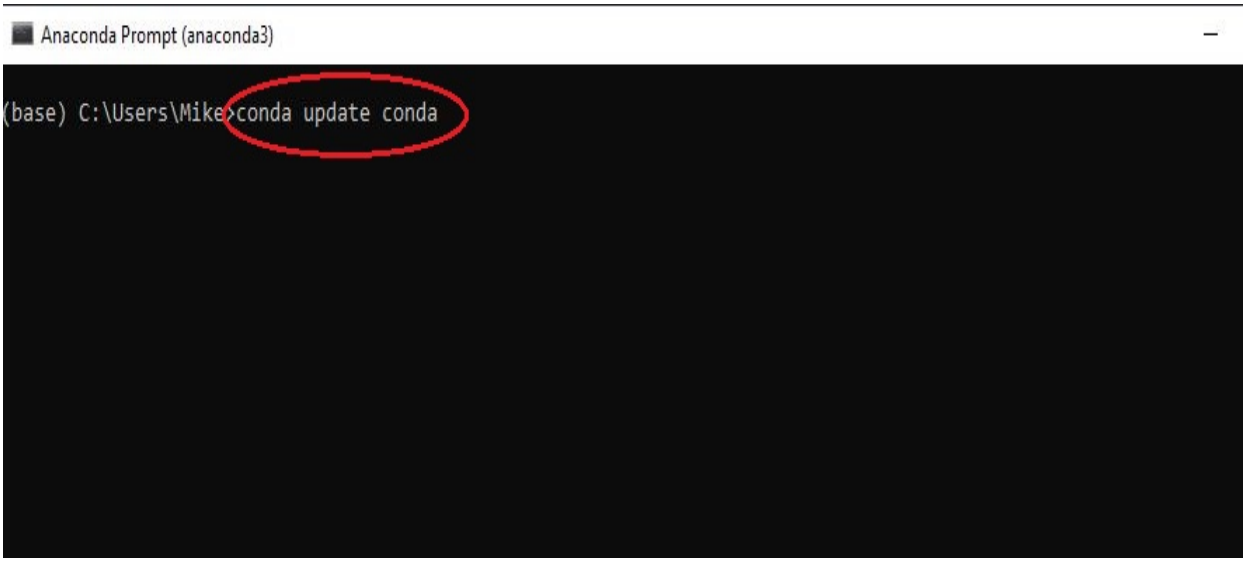


After clicking Anaconda3, you'll find Anaconda Prompt.

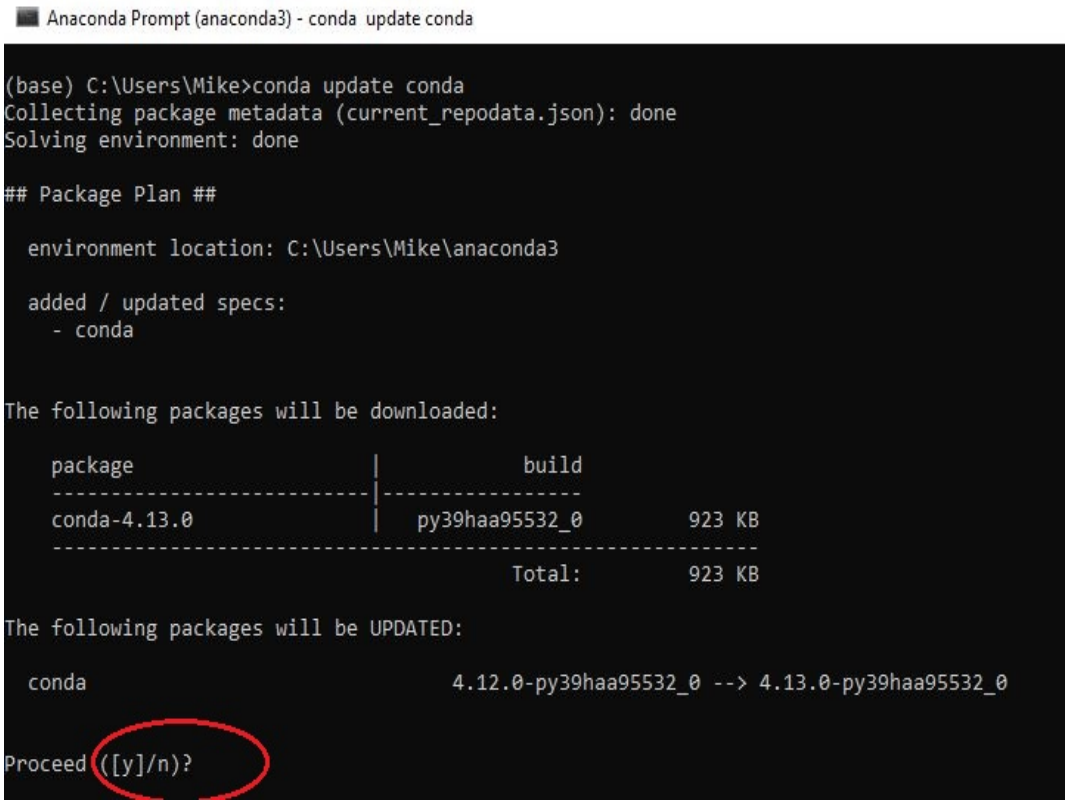


After clicking Anaconda Prompt, you'll open command-line window. In your system's command-line window, execute the following commands to update Anaconda's installed packages to their latest versions:

conda update conda



Following updating step by selecting **y**

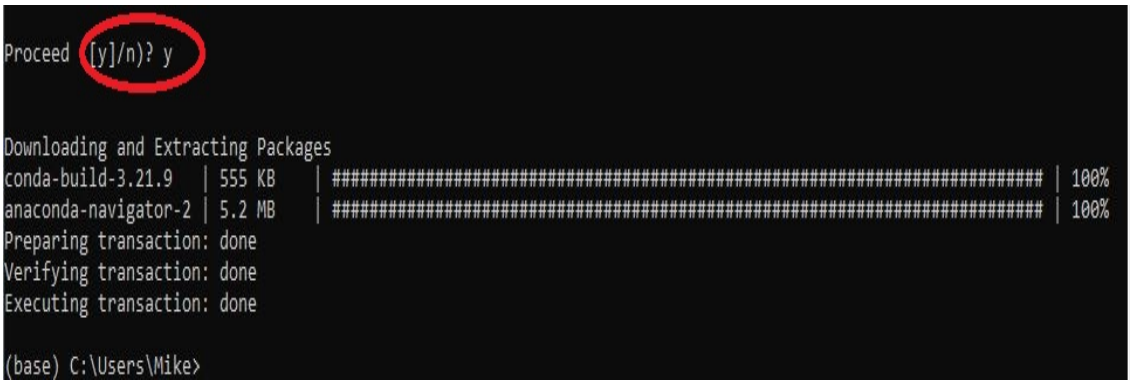


Then you need to run the following command:

conda update --all



Following updating step by selecting **y**



(7) Installation of Related Packages

We use two key Python package managers: **conda** package manager (used for conda command, and **pip** manager. Packages contain the files required to install a given Python library or tool.

(a) Installing Prospector Static Code Analysis Tool

In this course, we the Prospector analysis tool to analyze Python code. This tool checks Python code for common errors and help us improve code.

To install Prospector and the Python libraries it uses, you need to run the following command in the command-line window:

pip install prospector



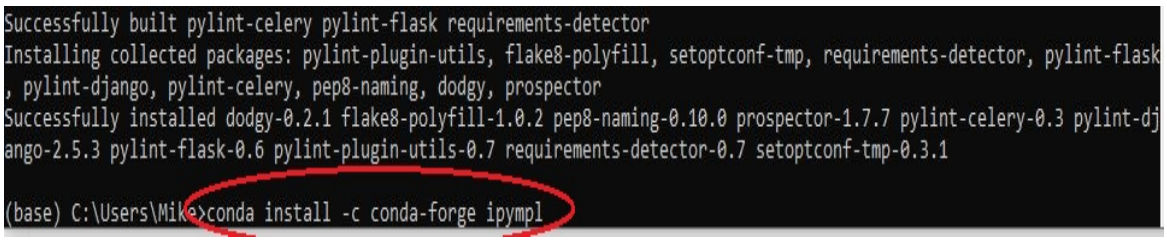
(b) Installing jupyter-matplotlib

Please note that if you download the Miniconda Installer software, you cannot complete this experiment (b).

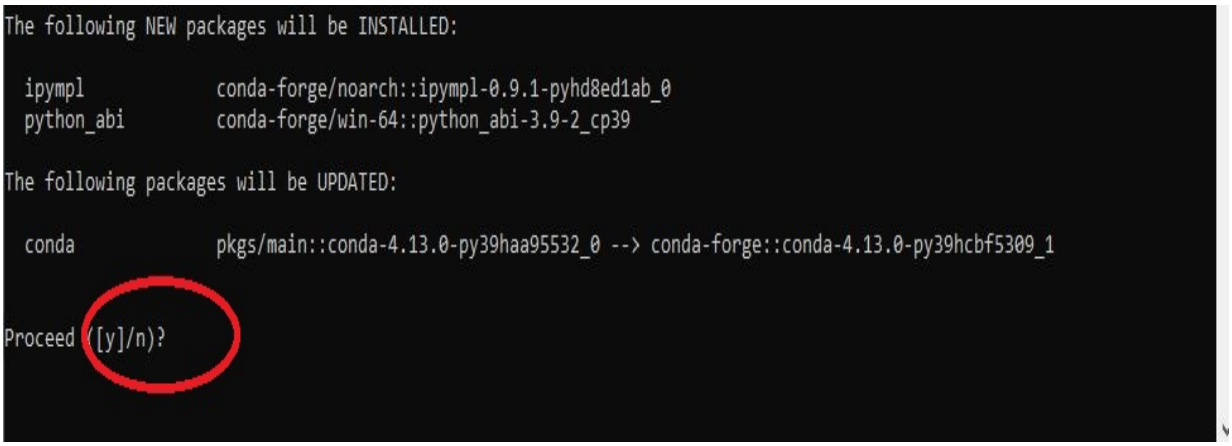
We use a visualization library called **Matplotlib** to show some animations. To use it in Jupyter Notebooks, you need to install a tool called ipympl.

Run the following commands in Anaconda Command Prompt

conda install -c conda-forge ipympl

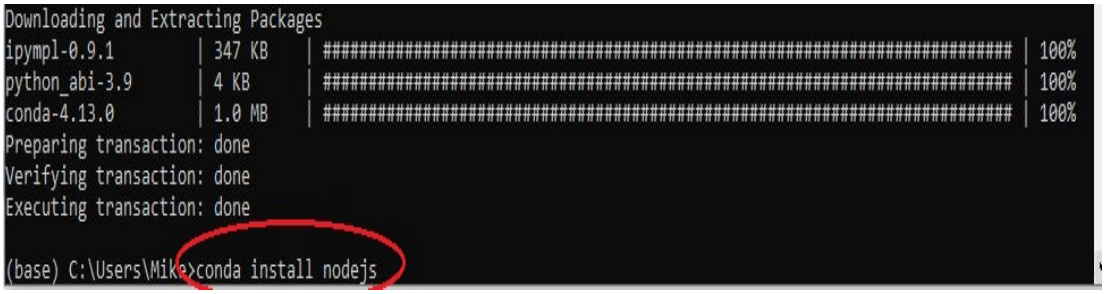


Then select y to continue:



Then Run the following commands in Anaconda Command Prompt

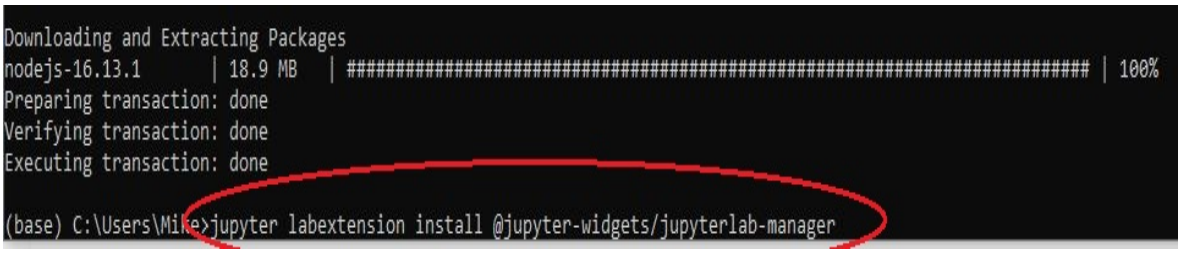
conda install nodejs





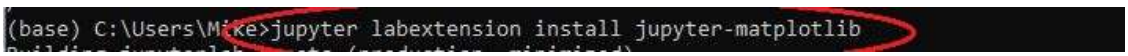
Then Run the following commands in Anaconda Command Prompt

jupyter labextension install @jupyter-widgets/jupyterlab-manager



Run the following commands in Anaconda Command Prompt

jupyter labextension install jupyter-matplotlib



3. Set up a local HTTP server

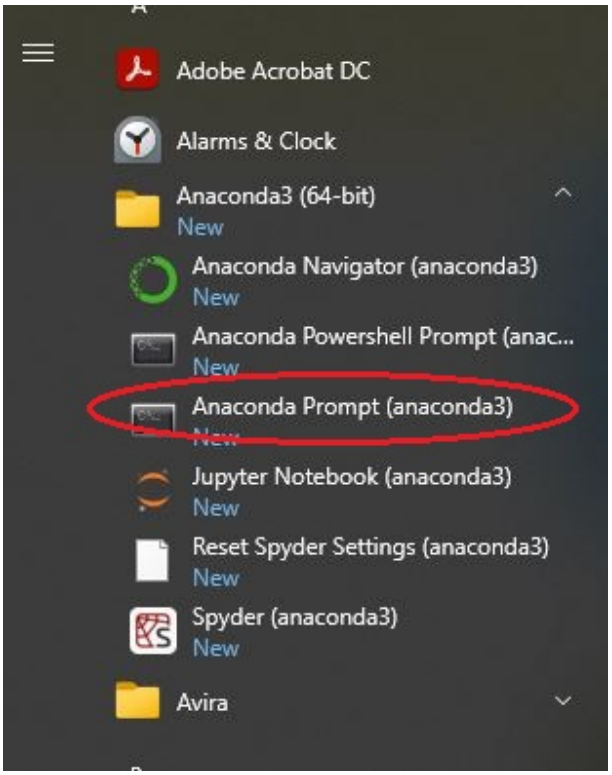
In the previous examples, we view local HTML files directly in our web browser. However, some browsers have restrictions that prevent them from loading local files via

JavaScript, for security reasons. It's a browser feature that prevents loading of scripts and other external files from untrusted third-party websites.

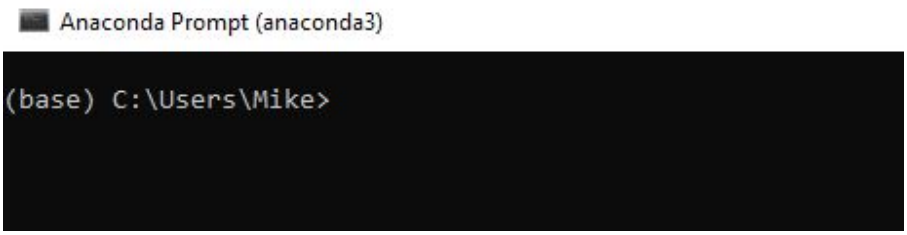
For this reason, it is more reliable to load your page via a web server. It is much faster to store and host everything locally (meaning, on the same computer). The idea is that the browser program requests files from the server program, and your server program responds by serving them back.

To run a Python web server:

(1) Open a new terminal window.



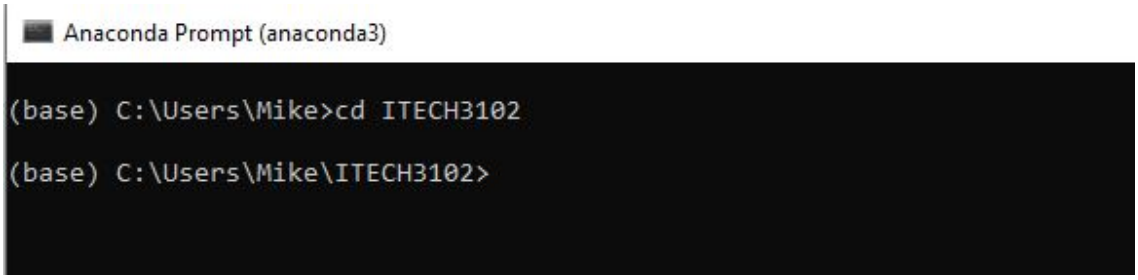
After clicking Anaconda Prompt, you'll open command-line window.



(2) Navigate to your project folder

As my project folder is ITECH3102, I use **cd** command (change directory) to change directory from the current C:\Users\Mike to working directory

C:\Users\Mike\ITECH3102.



(3) Run the following command:

python -m http.server 8888



This will activate the server on port 8888.

(4) View Web page from the server

Switch back to your web browser (do not close command-line window above) and type the following URL: http://localhost:8888/

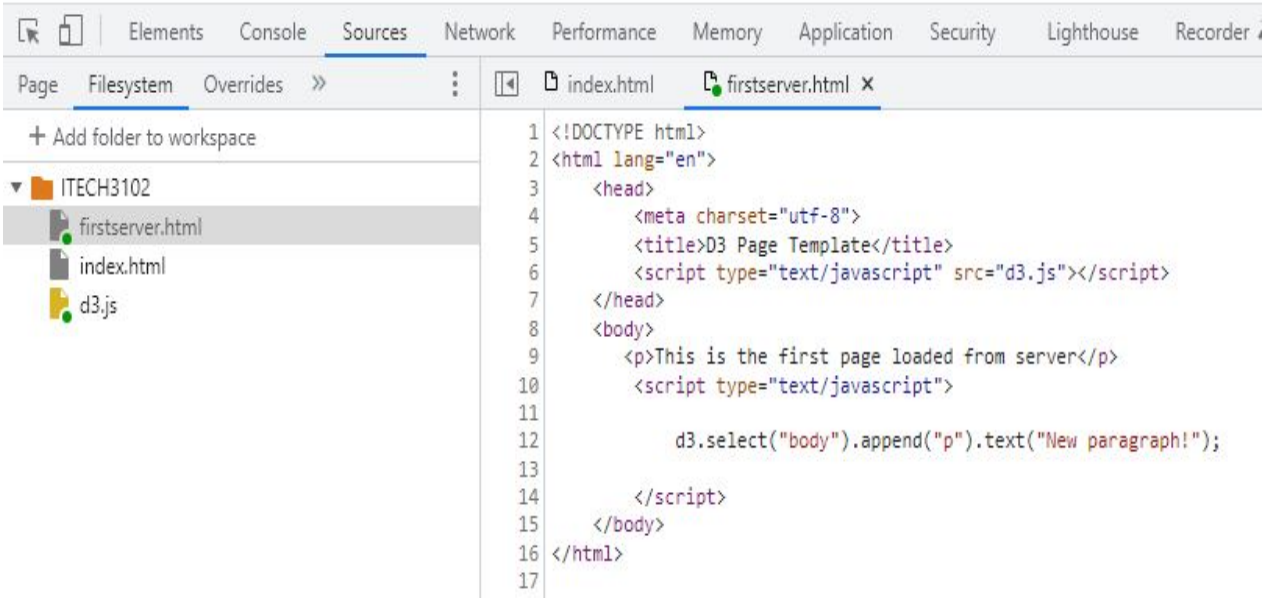


After typing the following URL: http://localhost:8888/ and pressing Enter key, the web server will generally serve a default page, which is often referred to "index" page (in our

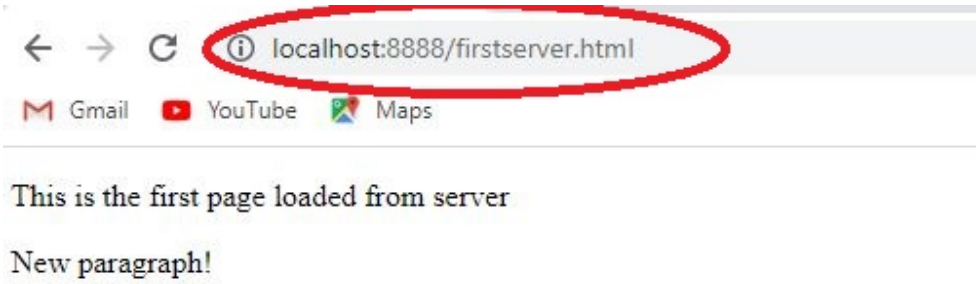
example, index.html). As there is no text in our index.html, browser will display empty page.

Now let us **to add some texts in html** and save it in the project folder\ITECH3102. Here I just copy the index.html template, add some codes in it and rename it as firstserver.html (so easy!).





Load firstserver.html page **via a web server** by typing the following URL: http://localhost:8888/firstserver.html.



We explain the above codes later.

If not viewed through a web server, the URL path will start with file:/// C:/Users/Mike/ITECH3102/firstserver.html. Confirm that the URL does not look like this:



Project 4. Answering questions

(1) Based on concepts of the marks and channels, provide some examples of appropriate and inappropriate encoding choices.

(2) Take an example of good visualizations (the graph presented in the topic 1, Napoleon's Army During the 1812, Visualization created by Charles Joseph Minard), and explain what marks and channel are used in the visualization, and discuss why they are used for?

Project 5. Creating a Professional Report

Summarize the above experiments procedure, results, answering questions and screenshots (project 1, 2, 3, 4) into one report. Your report is the assignment that is required to be submitted for evaluation on week 4. Create a report by following below steps.

Please note that

- as you may have a different version of Microsoft Word, the **following steps are just for your references.**

You can add a chapter called Chapter 4 in your previous report.

1. Open your last week's report and find the end of last week's report.
2. Copy this week's related experimental results, your findings and Screenshots, and paste them at the end of last week's report.
3. Delete original Table of Content you created.
4. Select all content, align all text to both left and right margin
5. Use shortcut key approach, generate Chapter 4: **SVG, D3**
6. Then use shortcut key approach to generate proper sub-chapters for this week's lab work.
7. Insert Table of Contents to your report.